

Solutions to Worksheet #2

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1. **Solution.** We need the coefficient of $m_{(3^2, 1^2)}$ in

$$e_{(2^2, 1^4)} = e_2^2 e_1^4.$$

It is enough to find the coefficient of $x_1^3 x_2^3 x_3 x_4$. The two labeled copies of e_2 choose two-element subsets $A, B \subseteq \{1, 2, 3, 4\}$. After these choices, let $r = (r_1, r_2, r_3, r_4)$ be the exponents still needed from the four labeled copies of e_1 . Those four factors can supply r in $4!/(r_1! r_2! r_3! r_4!)$ ways. Grouping the possible ordered pairs (A, B) by r gives

r	$\#(A, B)$	$4!/(r_1! r_2! r_3! r_4!)$	contribution
(1, 1, 1, 1)	1	24	24
(1, 2, 0, 1)	2	12	24
(1, 2, 1, 0)	2	12	24
(1, 3, 0, 0)	2	4	8
(2, 1, 0, 1)	2	12	24
(2, 1, 1, 0)	2	12	24
(2, 2, 0, 0)	6	6	36
(3, 1, 0, 0)	2	4	8

The total is

$$M_{(2^2, 1^4), (3^2, 1^2)}(e, m) = 172.$$

2. **Solution.** The coefficient of m_{1^n} in e_λ is the coefficient of the square-free monomial $x_1 x_2 \cdots x_n$. If $\lambda = (\lambda_1, \dots, \lambda_\ell)$, the factor e_{λ_i} must choose a subset of λ_i of the n variables, and the chosen subsets must be disjoint. Thus we partition the n labeled variables into labeled blocks of sizes $\lambda_1, \dots, \lambda_\ell$. The number of choices is the multinomial coefficient

$$M_{\lambda, 1^n}(e, m) = \binom{n}{\lambda_1, \dots, \lambda_\ell} = \frac{n!}{\lambda_1! \cdots \lambda_\ell!}.$$

3. **Solution.** The coefficient of m_n is the coefficient of x_1^n . Every monomial in e_r is squarefree, so one factor e_r can contribute at most one copy of

x_1 . We can obtain x_1^n only when $e_\lambda = e_1^n$, and then there is exactly one choice. Therefore

$$M_{\lambda,n}(e, m) = \begin{cases} 1, & \lambda = 1^n, \\ 0, & \text{otherwise.} \end{cases}$$

4. **Solution.** Assume $n \geq 2$ and consider $x_1^{n-1}x_2$. Since every factor e_{λ_i} is squarefree, there must be at least $n-1$ factors. A partition of n having at least $n-1$ parts is either 1^n or $(2, 1^{n-2})$.

If $\lambda = 1^n$, exactly one of the n labeled factors e_1 chooses x_2 , so the coefficient is n . If $\lambda = (2, 1^{n-2})$, the factor e_2 must choose x_1x_2 and all other factors choose x_1 , giving coefficient 1. Hence

$$M_{\lambda,(n-1,1)}(e, m) = \begin{cases} n, & \lambda = 1^n, \\ 1, & \lambda = (2, 1^{n-2}), \\ 0, & \text{otherwise.} \end{cases}$$

5. **Solution.** The statement is false. Take

$$\lambda = (4, 1, 1), \quad \mu = (3, 3).$$

Both are partitions of 6, and $\lambda >_{\text{lex}} \mu$ because $4 > 3$. Their conjugates are

$$\lambda' = (3, 1, 1, 1), \quad \mu' = (2, 2, 2).$$

Now $\mu' <_{\text{lex}} \lambda'$, rather than $\mu' >_{\text{lex}} \lambda'$. This is a counterexample.

6. **Solution.** Let

$$E(t) = \prod_{i \geq 1} (1 + x_i t) = \sum_{r \geq 0} e_r t^r$$

and let ω be a primitive cube root of unity, so $1 + \omega + \omega^2 = 0$. Since

$$1 + x + x^2 = (1 - \omega x)(1 - \omega^2 x),$$

we obtain

$$\prod_{i \geq 1} (1 + x_i + x_i^2) = E(-\omega)E(-\omega^2) = \left(\sum_{r \geq 0} (-\omega)^r e_r \right) \left(\sum_{s \geq 0} (-\omega^2)^s e_s \right).$$

This expression has integer coefficients because the two factors are complex conjugates. For example, its first homogeneous pieces are

$$1 + e_1 + (e_1^2 - e_2) + (e_1 e_2 - 2e_3) + \cdots$$

7. **Solution.**

- (a) Only the first five rows and first three columns can be nonzero. The row sums are $(3, 1, 1, 1, 1)$ and the column sums are $(4, 2, 1)$. The row of sum 3 must be $(1, 1, 1)$. After removing it, the four remaining rows each contain one 1, and the remaining column sums are $(3, 1, 0)$. Choose which three of the four rows place their 1 in the first column; the remaining row uses the second column. Thus

$$M_{(4,2,1),(3,1^4)}(e, m) = \binom{4}{3} = 4.$$

- (b) Transposition is a bijection. A matrix with row sums μ_i and column sums λ_j is sent by transposition to a matrix with row sums λ_j and column sums μ_i . Therefore

$$M_{\lambda, \mu}(e, m) = M_{\mu, \lambda}(e, m).$$

- (c) In the tableaux interpretation, row j of the Young diagram of λ is filled with distinct positive integers, written in increasing order, and integer i occurs μ_i times. Define $a_{ij} = 1$ when integer i occurs in tableau row j , and define $a_{ij} = 0$ otherwise. Distinctness within a tableau row makes this a $\{0, 1\}$ -matrix. Its row sum is μ_i and its column sum is λ_j .

Conversely, from such a matrix, fill tableau row j with the indices i for which $a_{ij} = 1$, in increasing order. These constructions are inverse to one another, so they give the required bijection.

8. **Solution.** The coefficient counts weakly increasing fillings of rows of lengths $(2, 2, 1, 1, 1, 1)$ with content $(3, 3, 1, 1)$. Equivalently, it counts nonnegative-integer matrices with those column and row sums.

Choose the contents A, B of the two labeled rows of length 2. The four rows of length 1 then supply a residual vector r . As in Exercise 1, their contribution is $4!/(r_1!r_2!r_3!r_4!)$. Grouping the possible pairs (A, B) gives

r	$\#(A, B)$	contribution
$(0, 2, 1, 1)$	2	24
$(0, 3, 0, 1)$	2	8
$(0, 3, 1, 0)$	2	8
$(1, 1, 1, 1)$	3	72
$(1, 2, 0, 1)$	4	48
$(1, 2, 1, 0)$	4	48
$(1, 3, 0, 0)$	4	16
$(2, 0, 1, 1)$	2	24
$(2, 1, 0, 1)$	4	48
$(2, 1, 1, 0)$	4	48
$(2, 2, 0, 0)$	6	36
$(3, 0, 0, 1)$	2	8
$(3, 0, 1, 0)$	2	8
$(3, 1, 0, 0)$	4	16

Adding the contributions yields

$$M_{(2^2, 1^4), (3^2, 1^2)}(h, m) = 412.$$

9. **Solution.** To obtain $x_1 x_2 \cdots x_n$ from $h_\lambda = h_{\lambda_1} \cdots h_{\lambda_\ell}$, the variables assigned to different factors must be disjoint. Once a subset of size λ_i is assigned to a factor, its variables occur in one weakly increasing order. The same multinomial count as in Exercise 2 therefore gives

$$M_{\lambda, 1^n}(h, m) = \frac{n!}{\lambda_1! \cdots \lambda_\ell!}.$$

10. **Solution.** Each h_{λ_i} contains $x_1^{\lambda_i}$ with coefficient 1. Choosing this term from every factor is the unique way to obtain x_1^n . Thus

$$M_{\lambda, n}(h, m) = 1 \quad \text{for every } \lambda \vdash n.$$

11. **Solution.** To obtain $x_1^{n-1} x_2$, exactly one factor h_{λ_i} must contribute $x_1^{\lambda_i-1} x_2$; every other factor contributes a pure power of x_1 . Each factor can be selected in exactly one way. Therefore

$$M_{\lambda, (n-1, 1)}(h, m) = \ell(\lambda),$$

where $\ell(\lambda)$ is the number of parts of λ .

12. **Solution.** Write

$$p_\lambda = p_{\lambda_1} \cdots p_{\lambda_\ell}, \quad p_{\lambda_j} = \sum_{i \geq 1} x_i^{\lambda_j}.$$

Choosing $x_i^{\lambda_j}$ from the j -th factor is equivalent to filling every cell of row j of the Young diagram of λ with the constant entry i . The exponent of x_i in the resulting monomial is the total number of cells containing i . Thus the monomial has type μ exactly when integer i occurs μ_i times. This proves the stated interpretation.

13. **Solution.** There are two labeled rows of length 2 and four labeled rows of length 1. To obtain content $(3, 3, 1, 1)$, the two rows of length 2 must receive different labels, namely 1 and 2. This can be done in $2!$ ways. The four rows of length 1 must then receive labels 1, 2, 3, 4, one each, in $4!$ ways. Hence

$$M_{(2^2, 1^4), (3^2, 1^2)}(p, m) = 2! 4! = 48.$$

14. **Solution.** In a constant-row filling with content 1^n , each row must have length 1. Thus the coefficient is zero unless $\lambda = 1^n$. If $\lambda = 1^n$, the n labeled one-cell rows may receive the n distinct labels in $n!$ ways. Therefore

$$M_{\lambda, 1^n}(p, m) = \begin{cases} n!, & \lambda = 1^n, \\ 0, & \text{otherwise.} \end{cases}$$

15. **Solution.** There is one constant-row filling of content (n) : fill every row with 1. Equivalently, every factor p_{λ_i} contributes $x_1^{\lambda_i}$. Hence

$$\boxed{M_{\lambda,n}(p, m) = 1} \quad \text{for every } \lambda \vdash n.$$

16. **Solution.** To obtain content $(n-1, 1)$, one row of length 1 must be filled with 2 and every other row must be filled with 1. Any part of λ equal to 1 can be chosen for that row. Consequently,

$$\boxed{M_{\lambda,(n-1,1)}(p, m) = m_1(\lambda)},$$

where $m_1(\lambda)$ denotes the multiplicity of the part 1 in λ .